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Stage-specific impacts of digital technologies on supply chain resilience: meta-analytic evidence for continuous process improvement

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Abstract

Purpose – This study quantifies how digital technologies enhance business process resilience across supply chain stages, addressing a critical gap in process management literature on technology-enabled continuous improvement.

Design/methodology/approach – A random-effects meta-analysis of 70 studies (218 effect sizes) maps digital technologies (sensing, analytics, collaboration) to three supply chain resilience process stages (preparedness, response and recovery), with moderator analysis of economic and cultural factors.

Findings – Digital technologies are positively associated with resilience ($r = 0.331$), preparedness ($r = 0.332$) and response ($r = 0.322$) gain more than recovery ($r = 0.273$). Analytics tools yield largest benefits in early stages ($r = 0.36$); recovery processes rely more on sensing technologies ($r = 0.299$). Effects are stronger in developing economies ($r = 0.350$) than developed ones ($r = 0.267$) and diminish as digital maturity rises ($r = -0.007$). Notably, long-term orientation culture ($r = 0.003$) positively moderates technology effectiveness, demonstrating how cultural factors shape digital transformation trajectories in business processes.

Practical implications – Organizations should adopt stage-specific technology deployment strategies while balancing technological implementation with governance mechanisms. In digitally mature environments, process managers should prioritize institutional improvements over additional technology investments to overcome diminishing returns and achieve sustainable resilience.

Originality/value – Our analysis reveals digital technologies more effectively support the "risk perception–dynamic response" process loop than "post-disruption reconfiguration–adaptive learning" processes. The identified "diminishing digital dividend" phenomenon challenges assumptions that technology alone ensures process resilience.

Keywords Supply chain resilience, Digital technologies, Meta-analysis, Business process management, Continuous improvement, Cultural moderation

Paper type Literature review

1. Introduction

In the face of unprecedented global uncertainty, managing business processes across supply chains has become a critical strategic imperative. Recent disruptions – including the COVID-19 pandemic, geopolitical conflicts and climate-related disasters – have exposed fundamental vulnerabilities in cross-organizational business process integration, highlighting the urgent need for resilience-building strategies (Zavala-Alcívar *et al.*, 2023). More recently, escalating China–US trade tensions have triggered profound reconfigurations of business processes, as rising protectionism and regionalization trends necessitate a reassessment of outsourcing and network design (Kazancoglu *et al.*, 2024). Within this challenging landscape, digital transformation offers promising pathways for business process managers. Research indicates that digital technologies can strengthen organizational responsiveness through advanced risk forecasting, real-time information processing and adaptive resource allocation (Ghobakhloo *et al.*, 2025).

As organizations embed digital tools into their process management systems (Ho *et al.*, 2023), these technologies have become recognized as essential enablers for improving how

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supply chains respond to, recover from and adapt to disruptions (Li *et al*., 2025). However, the business process management literature lacks a systematic, quantitative synthesis of how specific digital technologies differentially enhance supply chain resilience (SCR) across various operational contexts. Three critical gaps remain unaddressed:

First, resilience is often treated as a single, uniform concept, leading to the misleading assumption that technology enhances it in the same way across all situations. This view is problematic because SCR comprises distinct capabilities activated at different stages of a disruption (Stadtfeld and Gruchmann, 2024). For instance, the capacity to absorb a shock is different from the capacity to adapt and restore operations. The value of digital technologies, therefore, depends on the specific functions they support, from basic data visibility to more advanced capabilities like strategic flexibility and adaptive learning (Ghobakhloo *et al*., 2025). A "one-size-fits-all" approach to technology thus fails to capture the distinct value that digital tools provide at each process stage (Al-Swidi *et al*., 2024). Second, as the technological landscape evolves, research typically investigates either individual technologies or specific business contexts, limiting our comprehensive understanding of how diverse digital solutions contribute to resilience-building across different process dimensions (Tortorella *et al*., 2024). For instance, some researchers have stratified Industry 4.0 technologies into foundational, advanced and adaptive layers using the Delphi method (Spieske *et al*., 2023), highlighting the multifaceted nature of the digital technology spectrum in business process environments. Third, from a process management perspective, the relationship between digital technologies and SCR exhibits nonlinear characteristics embedded within complex socioeconomic and cultural contexts (Zhao *et al*., 2023). Research indicates that macro-level factors – including economic development indicators and cultural dimensions – exert significant moderating effects on organizations' digital adoption patterns and business process risk management approaches (Rodríguez-González *et al*., 2023).

While existing reviews have confirmed a generally positive technology–resilience relationship (e.g. Iftikhar *et al*., 2024; Spieske and Birkel, 2021), none have quantitatively deconstructed this relationship to reveal stage-specific effect sizes, leaving practitioners without empirical guidance for targeted technology deployment. This study is the first meta-analysis to provide comprehensive quantitative evidence of how the impact of digital technologies varies systematically across the different stages of resilience. In doing so, it challenges the common but untested assumption that technology provides uniform benefits. This oversight has left managers and scholars without stage-specific evidence, a critical gap this research aims to fill.

To address these research gaps – specifically, the need to (1) disaggregate the unitary construct of resilience into distinct process stages, (2) synthesize fragmented evidence on how different digital technologies perform across these stages and (3) quantitatively assess the influence of macro-level contextual factors – this meta-analysis is guided by four fundamental questions. RQ1 first maps the technological landscape. Subsequently, RQ2 and RQ3 directly tackle the stage-specific and technology-specific effects, while RQ4 examines the critical economic and cultural moderators. Specifically, we ask:

RQ1. Which digital technologies feature prominently in SCR-related business process research?

RQ2. What are the quantitative effect sizes of digital technologies on SCR overall, and on each of its three operational stages in organizational contexts?

RQ3. Which moderating variables influence how digital technologies enable SCR in business process systems?

RQ4. From a global business perspective, how do economic and cultural factors shape this technological relationship?

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This meta-analysis makes several contributions to business process management research. Theoretically, by examining the three critical SCR phases – preparedness, response and recovery – and systematically cross-analyzing them with different categories of digital technologies, our work transcends the limitations of qualitative studies and single-dimensional empirical analyses. The resulting framework offers practical, stage-specific guidance for applying digital technologies, supporting a shift from isolated process reengineering to continuous improvement. From a practical standpoint, our findings deliver evidence-based decision support for business process managers and policymakers, enabling organizations to identify optimal digital approaches for enhancing process resilience at different operational stages and offering targeted strategies for managing cross-organizational processes under varying economic and cultural contexts.

2. Literature review and early work

2.1 Resilience and technology in supply chains

SCR – a network's dynamic capability to rebound from disruptions – has been a central theme in operations research since Christopher and Peck (2004) introduced the concept. Drawing on publication data from the Web of Science SSCI and SCI databases, Figure 1 confirms scholarly interest in SCR has grown exponentially since 2015. This surge mirrors both the increasing frequency of global shocks and the rapid diffusion of digital technologies (Stadtfeld and Gruchmann, 2024).

Digitalization has advanced in two discernible waves. During 2014–2020 firms experimented with isolated tools, gradually aligning them with resilience routines. Since 2021 – accelerated by COVID-19 – IoT sensors, big-data analytics, blockchain, digital twins and, from 2022, advanced AI have converged into an end-to-end ecosystem that supports real-time monitoring, risk assessment and adaptive decision-making (Li *et al.*, 2025; Spieske and Birkel, 2021). Collectively these tools heighten visibility, collaboration, agility and responsiveness, enabling earlier disruption detection and faster recovery (Iftikhar *et al.*, 2024). By dissolving conventional boundaries among information, talent and capital flows, they embody Industry 4.0 principles of modularity, interoperability and decentralization (Büyüközkan and Göçer, 2018; Ghobakhloo *et al.*, 2025).

Regarding how SCR is segmented, multiple research perspectives exist: Wieland and Wallenburg (2013) classify resilience from an attribute perspective into passive resilience (agility) and active resilience (robustness); Pillay (2017), approaching from capability

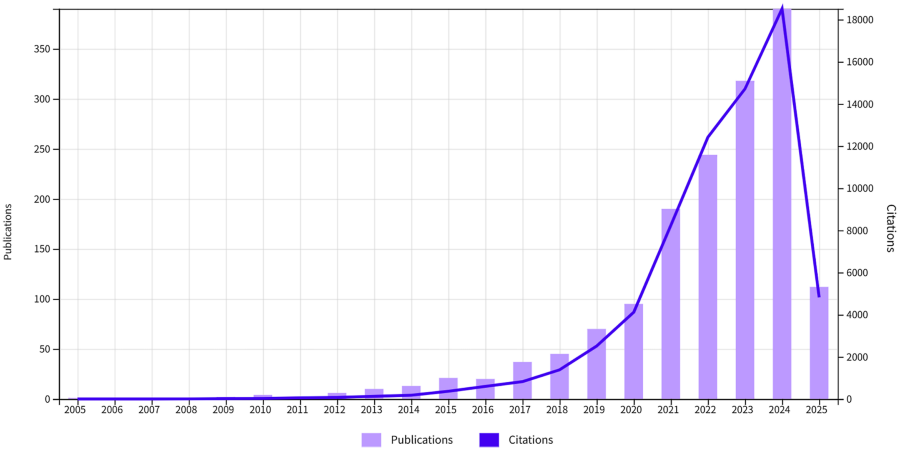


Figure 1. Publication and citation trends in supply chain resilience research. Source: Authors' own work

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building, proposes four key competencies – response, monitoring, learning and foresight. Other scholars have mapped SCR to specific functional elements such as production/manufacturing, logistics/distribution, demand management and collaboration and returns/remanufacturing (Akçalı and Çetinkaya, 2011; Jung *et al.*, 2014).

While various frameworks exist, this study adopts the process-oriented three-stage classification from Ponomarov and Holcomb (2009), as it aligns with a business process lifecycle view of disruption management. This model conceptualizes resilience not as a single outcome, but as a dynamic capability unfolding across distinct phases. The first is preparedness, which involves proactive capabilities for "event readiness", such as developing risk warning systems and establishing resource redundancy. This is followed by response, centered on executing an "efficient response" to a disruption through actions like flexible resource deployment and innovative decision-making. The final stage is recovery, which extends beyond simple restoration to include system reconfiguration and adaptive learning, enabling the supply chain to return to its original state or evolve to a better one. This granular, stage-based operationalization, detailed in Table 1, forms the analytical foundation for quantifying the differentiated impact of digital technologies.

2.2 Overview of prior review literature

Early reviews focused on defining resilience and its metrics (Hosseini *et al.*, 2019). More recently, attention has shifted to the digitalization–resilience nexus, with reviews showing that Industry 4.0 enablers elevate disruption tolerance (Spieske and Birkel, 2021) and that technologies like digital twins can enhance real-time scheduling (Golan *et al.*, 2021). However, it is also argued that genuine resilience gains require cross-organizational collaboration, not just technology (Ghobakhloo *et al.*, 2025).

Despite this progress, critical gaps remain, which this study addresses. Most reviews treat SCR as a unitary construct, obscuring how digital tools perform across preparedness, response and recovery (Spieske and Birkel, 2021). Furthermore, evidence is often fragmented, and large-sample quantitative validation is scarce.

Our study overcomes these limitations with a random-effects meta-analysis that (1) maps digital technologies to the three SCR stages, (2) quantifies their stage-specific effect sizes and (3) tests macro-level moderators. This approach advances SCR theory with robust statistical evidence and offers managers stage-aware guidance for technology deployment, reinforcing the strategic value of digital transformation.

3. Research design

Following Ponomarov and Holcomb's (2009) three-stage model categorizing SCR into preparedness, response and recovery, this study employs meta-analysis guidelines by Pigott and Polanin (2020) to assess how digital technologies impact these stages differently.

3.1 Literature search and inclusion criteria

Adhering to PRISMA guidelines, our systematic literature search queried four major databases (Web of Science, Scopus, ProQuest and Emerald) to identify all relevant studies published, or made available via early access, between January 2004 and April 2025. The search strategy intentionally focused on resilience concepts – using a representative query such as TS= ("supply chain resilience" OR "supply chain robustness" OR "supply chain adaptation") – rather than specific technology keywords. This approach was designed to ensure a broad and unbiased capture of diverse and evolving digital solutions.

The search and screening protocol is detailed in Figure 2. The initial search yielded 8,427 records. After removing 5,371 duplicates, we screened the titles and abstracts of the remaining 3,056 articles, excluding 2,057 that did not meet the research topic (e.g., they were non-English, review articles or outside the business process scope). A full-text review of the 999

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Table 1. SCR four-stage coding

Stage	Indicator name	Definition/explanation
Preparedness	Risk Warning	Systematically identifies, evaluates and flags potential risks, establishing a forward-looking prevention foundation for the supply chain
	Information Coordination	Achieves full-chain information transparency and integration, improving communication and decision-making efficiency across nodes
	Resource Redundancy	Builds buffers through redundant resources and robust infrastructure to safeguard against external shocks
	Organizational Learning	Enhances overall risk response and prevention capabilities through skill development, network governance and the accumulation of experience
Response	Flexible Response	Rapidly reallocates and integrates both internal and external resources to enable swift adjustments and responses under disruption conditions
	Innovative Decision-Making	Leverages improvisational innovation and contextual judgment in dynamic environments to drive precise decisions and plan execution
	Disruption Containment	Implements effective measures to isolate disruptions and prevent their spread, ensuring core operations and services remain uninterrupted
Recovery	Speed of Restoration	Measures how quickly key functions return to normal operation post-disruption, reflecting the system's self-repair capacity
	Reconfiguration	Undertakes structural reorganization and process redesign based on disruption experiences, promoting optimized realignment of the supply chain as a whole
	Resource Regeneration	Enables swift replenishment and regeneration of strategic resources, restoring supply and production capacity balances disrupted by interruptions
	Evolution and Upgrading	Propels strategic adaptation and capability enhancements through experiential learning and continuous improvement, fostering long-term competitive advantages
Combined	Systemic Resilience	Integrates resilience-building achievements from all stages, reflecting the overall capacity of the supply chain to withstand shocks and maintain continuous operations
	Collaborative Integration	Reinforces cross-organizational and cross-boundary synergy, achieving resource sharing and risk-sharing to strengthen overall resilience
	Strategic Adaptation	Relies on dynamic adaptation and innovation, transforming resilience into sustainable competitive advantages, serving as a long-term market strategy foundation
	Benefit Realization	Maximizes the comprehensive economic and operational return on investments in resilience and governance optimization through strategic resource allocation

Source(s): Authors' own work

eligible articles led to the exclusion of 450 papers for lacking empirical research design. For final inclusion, studies had to provide sufficient statistical data for effect-size calculation. This critical step led to the exclusion of 424 articles with no available data and a further 55 with insufficient data, yielding the final sample of 70 studies for meta-analysis.

3.2 Study characteristics and intervention coding

We established a multidimensional coding framework capturing:

(1) Study background (publication year, journal type, research method, economic development level, firm size, industry)

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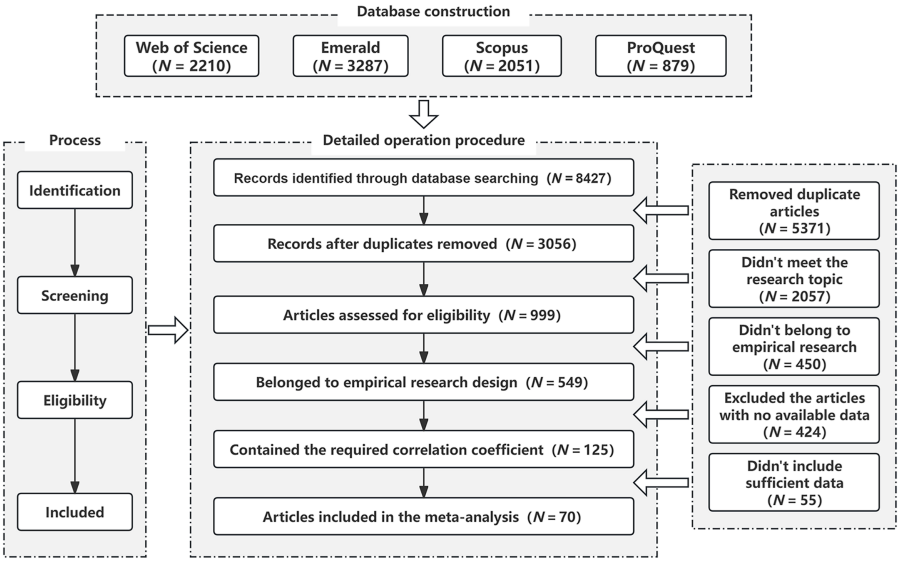


Figure 2. Search flow diagram. Source: Authors' own work

(2) SCR stage was coded into preparedness, response and recovery, plus a combined category for studies addressing multiple phases non-distinctly, based on the specific indicators in Table 1 which were refined through iterative discussion.

(3) Digital technology type (sensing/perception, analytics/decision-making, collaboration/integration, multiple/comprehensive)

A complete list of the 70 primary studies included in the meta-analysis is presented in Appendix A. Two researchers independently coded the studies; Cohen's kappa values for both SCR stage and technology classification exceeded 0.80, indicating high inter-rater reliability. Any disagreements were resolved through discussion until consensus was reached.

3.3 Sensitivity analysis and heterogeneity tests

Sensitivity analysis using a leave-one-out approach confirmed minimal fluctuation in effect sizes (interval [0.326, 0.334]), indicating resistance to outliers. Heterogeneity assessment via Cochran's Q test and the I^2 statistic showed significant heterogeneity across all subsamples ($I^2 > 90\%$), justifying our random-effects model and the reporting of the variance of true effect sizes (τ^2) (Table 2).

3.4 Publication bias

We assessed publication bias following Oesterreich *et al.* (2025) by examining funnel-plot symmetry, Egger's regression test and fail-safe analyses. The overall funnel plot (Figure A1) was essentially symmetrical, and Egger's test was non-significant ($p > 0.05$), indicating minimal small-study effects. The classic Rosenthal's (1979) fail-safe N reached 335,369 – far above the 5k + 10 threshold of 525 – and Orwin's (1983) fail-safe N of 6,430 to reduce the standardized mean difference to 0.01 further confirms the stability of our overall effect-size estimate.

Stage-specific funnel plots (Figure A2) appeared largely symmetrical; however, the more sensitive Egger's regression test revealed a significant publication bias for the response stage

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Table 2. Publication bias

Resilience phase	K	Egger's Score	SD	z	*p*	Fail-safe *N* Classic	Orwin–0.01
Readiness	69	−0.80	1.950	−0.41	0.6807	33,215	2,041
Response	55	−5.89	2.554	−2.30	0.0212	21,911	1,556
Recovery	42	−0.34	2.465	−0.14	0.8902	8,796	1,040
Comprehensive	52	−1.38	1.567	−0.88	0.3788	23,864	1,795
Overall	218	−1.53	1.028	−1.49	0.1372	335,369	6,430

Source(s): Authors' own work

($p < 0.05$). This suggests the reported effect size for this stage ($r = 0.322$) is best interpreted as a potential upper bound, as the bias likely inflates the estimate by under-representing studies with smaller or null findings. Nevertheless, the finding's statistical significance remains highly robust: the fail-safe *N* of 21,911 confirms that the positive relationship is highly unlikely to be nullified by this potential bias.

4. Results

4.1 Review of digital technologies in supply chain resilience (for RQ1)

Based on analysis of 70 empirical studies, this research categorizes digital technologies into three functional dimensions that form an integrated ecosystem for SCR development. Rather than classifying by chronology or type, we organize technologies by functionality:

4.1.1 Information acquisition and sensing technologies. Technologies such as the Internet of Things, radio frequency identification and sensors constitute the foundational layer for SCR, enabling continuous monitoring and transmission of supply chain status data. These technologies heighten visibility, allowing firms to identify potential risks and reduce reaction times to disruptions. As Huang *et al*. (2023) demonstrate, supply chain visibility mediates between information acquisition technologies and SCR, helping organizations recover faster. When paired with analytics, these technologies transition supply chains from passive defense to proactive warning systems.

4.1.2 Data analytics and decision-making technologies. Tools such as artificial intelligence (AI), machine learning, predictive analytics and digital twins transform data into actionable insights, which can enhance risk management efficiency. Belhadi *et al*. (2024) show that AI-driven innovations are linked to a boost in risk prediction capabilities in dynamic environments. These technologies provide forward-looking solutions through contingency optimization and adaptive supply network configurations. However, as Xu and Liu (2024) note, excessive reliance on analytics may diminish intuitive judgment, suggesting the need for balance between technological inputs and human insight.

4.1.3 Collaboration, integration and synchronization technologies. Blockchain, cloud computing, collaborative platforms and supply chain control towers represent the networked collaboration layer, dismantle information silos and facilitate connectivity among supply chain partners. These technologies are associated with enhanced SCR, partly by establishing unified information-sharing platforms, standardizing cross-organizational processes and increasing transparency. Harju *et al*. (2023) demonstrate that collaborative synergy and compatible standards are necessary to convert digital investments into resilience capabilities.

4.2 Primary effect-size calculations (for RQ2)

Using correlation coefficients (*r*) to measure effect sizes, this study employs the approach of Wang *et al*. (2024) to standardize other metrics into correlation coefficients, followed by

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Fisher's z-transformation via Equation (1), ensuring the accuracy and comparability of the estimated effect sizes. To enhance methodological transparency, Appendix B provides a detailed guide with concrete examples illustrating the conversion of commonly reported statistics (e.g. standardized regression coefficients) into Pearson's *r*. Following Cohen's (2013) guidelines, an effect size below 0.10 indicates a weak correlation, 0.10–0.30 indicates a moderate correlation and above 0.30 indicates a strong correlation.

$$z' = 0.5 \times \ln\left(\frac{1+r}{1-r}\right) \quad (1)$$

Meta-analysis results show that digital technologies have a strong positive association with SCR (r = 0.331, 95% CI [0.300, 0.363], $p < 0.001$) but display significant disparities among the three SCR stages – preparedness, response and recovery (Table 3). Specifically, the association is strongest for preparedness (r = 0.332), followed closely by response (r = 0.322), while the relationship is more moderate for recovery (r = 0.273). Of note, the combined stage exhibits a markedly higher effect size (r = 0.388), potentially stemming from synergistic effects when multiple technologies are applied jointly or because some studies did not isolate the recovery phase in their analyses.

4.3 Stage-specific impacts of digital technologies (for RQ3)

Further subgroup analysis reveals the interaction effects between technology type and resilience stage. Figure 3 provides a heatmap of these technology-stage synergies – where color intensity represents the effect size (*r*) – visually complementing the detailed statistical results in Table 4.

In the preparedness stage, data analytics and decision-making technologies are associated with the strongest effect size ($r = 0.361$), slightly surpassing that of collaboration technologies ($r = 0.355$), while sensing technologies show a lower but still significant effect ($r = 0.263$). This finding points to the importance of analytical capabilities in the processes of identifying latent threats and formulating preventive strategies.

For the response stage, analytics technologies maintain the strongest association ($r = 0.366$), followed by collaboration technologies ($r = 0.348$) and sensing technologies ($r = 0.314$). This distribution reflects organizations' reliance on data-driven decisions and resource deployment during disruptions.

In the recovery stage, sensing technologies (r = 0.299) are linked to a slightly stronger effect size than collaboration technologies (r = 0.290), while analytics technologies show a relatively lower effect (r = 0.256). This pattern suggests that processes related to real-time monitoring are particularly salient for reconfiguring supply chains and repairing damaged nodes.

Table 3. Random effects model for the influence of digital technologies on SCR across stages

Variable	k	g	SE	95% CI	Q	df	*p*	I^2 (%)	Tau-2	Tau
Readiness	69	0.332	0.029	[0.275, 0.389]	1099.31	68	0.000	93.21	0.054	0.232
Response	55	0.322	0.035	[0.253, 0.391]	879.52	54	0.000	94.26	0.063	0.251
Recovery	42	0.273	0.036	[0.202, 0.344]	715.79	41	0.000	94.90	0.050	0.224
Comprehensive	52	0.388	0.030	[0.330, 0.446]	568.57	51	0.000	90.24	0.040	0.200
Overall	218	0.331	0.016	[0.300, 0.363]	3804.92	217	0.000	93.94	0.053	0.230

Source(s): Authors' own work

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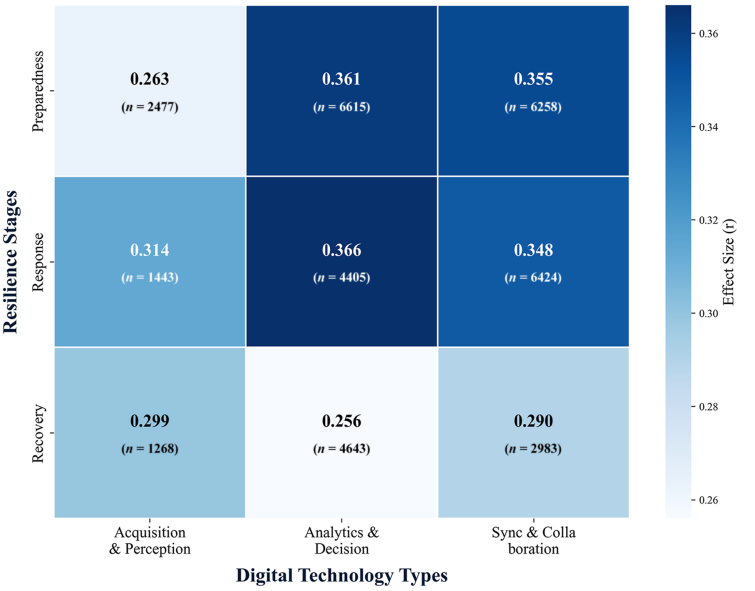


Figure 3. Heatmap of technology–stage synergies. Source: Authors' own work

4.4 Analysis of the moderating effects of technology's support for SCR (for RQ4)

Under the meta-analysis framework adopted in this study, subgroup analysis and meta-regression were conducted to examine whether the relationship between digital technologies and SCR varies across different contexts. An overview of both analyses is presented in Figure 4.

4.4.1 Subgroup analysis. As shown in Table 5, subgroup analyses were performed according to industry type, economic development level and firm size.

Industry type: The effect size for manufacturing ($g = 0.350$, 95% CI [0.300, 0.401]) is slightly higher than for services ($g = 0.322$, 95% CI [0.250, 0.394]), but this difference is not statistically significant ($p = 0.525$), indicating digital technologies' broad applicability across these categories. This broad classification, however, masks the nuanced role of technology in diverse sectoral contexts (e.g., humanitarian aid, construction) which, while having sample sizes too small for separate estimation, offer key insights. For instance, in humanitarian logistics, digital twin technology is shown to be a critical contributor to SCR (Singh, 2025), highlighting its value in highly volatile environments.

Economic development level: Developing economies show a significantly higher effect size ($g = 0.350$, 95% CI [0.314, 0.386]) than developed economies ($g = 0.267$, 95% CI [0.197, 0.337]) ($p = 0.039$). Regions with fewer legacy systems may find it easier to adopt emerging digital technologies and quickly strengthen SCR.

Firm size: Large enterprises ($g = 0.353$, 95% CI [0.305, 0.401]) and SMEs ($g = 0.300$, 95% CI [0.231, 0.370]) show no significant difference ($p = 0.219$). While larger organizations typically have more established resources, SMEs can leverage external partnerships and flexible digital tools to compensate for resource constraints.

4.4.2 Meta-regression. This study extends the subgroup analysis by using meta-regression to examine the macro-level moderating factors, specifically national economic development and cultural influences, that contextualize the relationship between digital technologies and SCR (see Equation (2)). The Digital Competitiveness Score is employed to measure a country's digital infrastructure maturity, while GDP per capita serves as a proxy for economic development. Additionally, several of Hofstede's cultural dimensions – such as long-term orientation and uncertainty avoidance, which are related to organizational practices – are

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Table 4. Effect sizes for digital technology–stage combinations (Fisher's z)

Resilience phase	Subsamples	*k*	*n*	Effect size (*r*)	95% CI	*Q* (*p*, χ^2)	I^2 (%)
Readiness	*Overall*	*69*	*17,901*	*0.332*	*[0.275, 0.389]*	*1099.31 (p = 0.000)*	*93.21*
	Acquisition and Perception	8	2,477	0.263	[0.088, 0.438]	182.98 (p = 0.000)	94.75
	Analytics and Decision	29	6,615	0.361	[0.295, 0.428]	190.19 (p = 0.000)	86.28
	Sync and Collaboration	20	6,258	0.355	[0.205, 0.506]	605.76 (p = 0.000)	97.19
	Comprehensive	12	2,551	0.265	[0.204, 0.325]	24.46 (p = 0.011)	54.99
Response	*Overall*	*55*	*14,585*	*0.322*	*[0.253, 0.391]*	*879.52 (p = 0.000)*	*94.26*
	Acquisition and Perception	5	1,443	0.314	[0.179, 0.449]	27.08 (p = 0.000)	84.91
	Analytics and Decision	19	4,405	0.366	[0.289, 0.444]	114.53 (p = 0.000)	85.07
	Sync and Collaboration	22	6,424	0.348	[0.206, 0.491]	601.10 (p = 0.000)	96.96
	Comprehensive	9	2,323	0.170	[0.047, 0.293]	65.71 (p = 0.000)	88.75
Recovery	*Overall*	*42*	*22,277*	*0.273*	*[0.202, 0.344]*	*715.79 (p = 0.000)*	*94.90*
	Acquisition and Perception	5	1,268	0.299	[0.151, 0.448]	29.83 (p = 0.000)	85.09
	Analytics and Decision	22	4,643	0.256	[0.148, 0.364]	290.26 (p = 0.000)	92.71
	Sync and Collaboration	11	2,983	0.290	[0.135, 0.444]	151.73 (p = 0.000)	94.48
	Comprehensive	4	13,383	0.275	[0.126, 0.424]	69.83 (p = 0.000)	92.91
Comprehensive stage	*Overall*	*52*	*12,143*	*0.388*	*[0.330, 0.446]*	*568.57 (p = 0.000)*	*90.24*
	Acquisition and Perception	11	2,729	0.361	[0.264, 0.457]	57.34 (p = 0.000)	83.24
	Analytics and Decision	17	3,464	0.361	[0.288, 0.435]	70.82 (p = 0.000)	78.55
	Sync and Collaboration	12	2,795	0.498	[0.317, 0.679]	313.35 (p = 0.000)	95.71
	Comprehensive	12	3,155	0.341	[0.252, 0.430]	68.65 (p = 0.000)	83.95
Grand mean (overall sample)	*Overall Sample*	*218*	*66,906*	*0.331*	*[0.300, 0.363]*	*3804.92 (p = 0.000)*	*93.94*
	Acquisition and Perception	29	7,917	0.313	[0.244, 0.382]	352.67 (p = 0.000)	89.3
	Analytics and Decision	87	19,127	0.366	[0.294, 0.378]	701.13 (p = 0.000)	88.36
	Sync and Collaboration	65	18,460	0.368	[0.289, 0.447]	1749.62 (p = 0.000)	96.60
	Comprehensive	37	21,412	0.269	[0.217, 0.320]	407.67 (p = 0.000)	89.24

Source(s): Authors' own work

incorporated. Previous cross-national research and meta-analytic studies underscore the significance of these macro and cultural variables in moderating technology adoption and risk

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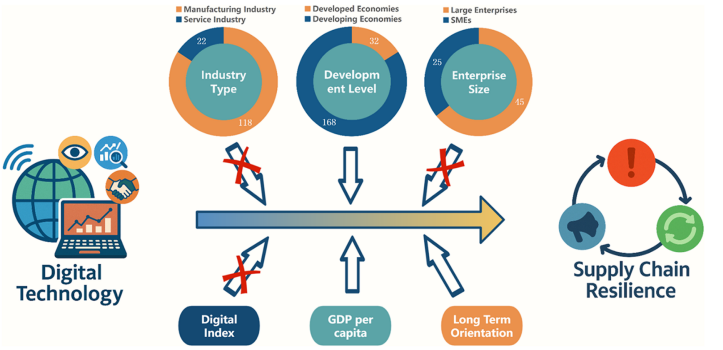


Figure 4. Moderating effects. Source: Authors' own work

Table 5. Subgroup analysis of moderating effects on supply chain resilience

Moderator	Subgroup	k	r	95% CI	*p*	I^2 (%)	Between *p* (χ^2)
Industry type (excluding mixed/ other)	Manufacturing Industry	118	0.350	[0.300, 0.401]	<0.001	94.79	0.525 (0.40)
	Service Industry	22	0.322	[0.250, 0.394]	<0.001	83.75	
	Overall	140	0.346	[0.302, 0.390]	<0.001	94.11	
Economic development level (excluding unreported)	Developed Economies	32	0.267	[0.197, 0.337]	<0.001	83.89	0.039 (4.24)
	Developing Economies	168	0.350	[0.314, 0.386]	<0.001	94.34	
	Overall	200	0.337	[0.305, 0.370]	<0.001	93.66	
Enterprise size (excluding mixed/ other)	Large Enterprises	45	0.353	[0.305, 0.401]	<0.001	87.42	0.219 (1.51)
	Small- and Medium-sized Enterprises (SMEs)	25	0.300	[0.231, 0.370]	<0.001	88.18	
	Overall	70	0.334	[0.295, 0.373]	<0.001	88.65	

Source(s): Authors' own work

management outcomes (Hofstede, 2001; Kumar and Raj, 2025). This approach thus examines how institutional, economic and cultural factors contribute to cross-national variations and assesses their role in moderating the incremental effects of digital technologies on SCR.

$$ES_{ij} = \beta_{0j} + r_{ij} \tag{2}$$

$$\begin{aligned}\beta_{0j} = \; & \gamma_{00} + \gamma_{01} n_j + \gamma_{02}\,\text{Year}_j + \gamma_{03}\,\text{DigitalIndex}_j + \gamma_{04}\,\text{GDPpc}_j \\ & + \gamma_{05}\,\text{PowerDistance}_j + \gamma_{06}\,\text{Individualism}_j + \gamma_{07}\,\text{Masculinity}_j \\ & + \gamma_{08}\,\text{UncertaintyAvoidance}_j + \gamma_{09}\,\text{LongTermOrientation}_j + \gamma_{010}\,\text{Indulgence}_j + u_{0j}\end{aligned} \tag{3}$$

where ES_{ij} is the ith effect size describing the relationship reported within the jth sample. Equation (3) describes the effect of all identified moderators that vary between studies. Here, u_j

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is the term for the study-level residual error. A fully nested model (Table 6) illustrates the meta-regression results, which provide additional insights.

Table 6 presents the fully nested meta-regression. The model explains 6.34% of the between-study variance in effect sizes ($R^2 = 0.0634$). Even after all moderators are included, the intercept remains positive and significant ($\gamma = 0.699$, $p < 0.01$), confirming a stable, positive link between digital technologies and SCR across contexts.

Moderator analysis, reported with standardized beta (β) coefficients, highlights three notable patterns. First, the digitalization index exerts a significant negative moderating effect ($\gamma = -0.007$, $p < 0.10$; $\beta = -0.1935$), indicating that the "digital dividend" diminishes as national digital infrastructure matures. Second, a culture's long-term orientation has a positive influence ($\gamma = 0.003$, $p < 0.10$; $\beta = 0.0709$), suggesting that forward-looking societies strengthen the technology–resilience relationship. Third, publication year shows a marginally negative coefficient ($\gamma = -0.020$, $p < 0.10$), hinting that reported effect sizes may be converging over time. No other cultural or economic indicators emerge as significant moderators.

Table 6. Meta-regression results (digital technology and supply chain resilience)

Parameter	Unstandardized Coeff. (γ)	SE	z-value	95% CI	Standardized (β)
Constants and controls					
Intercept (_cons)	0.6994***	0.230941	3.03	[0.2467147, 1.151987]	0.2955***
n (γ_{01})	−0.000013	0.0000155	−0.84	[−0.0000434, 0.0000173]	0.0048
year (γ_{02})	−0.0198*	0.0109648	−1.81	[−0.0413180, 0.0016631]	−0.0314*
Infrastructure – economy					
Digitalization Index (γ_{03})	−0.0073*	0.0038931	−1.88	[−0.0149429, 0.0003178]	−0.1935*
GDP per Capita (γ_{04})	3.81e−06	4.29e−06	0.89	[−4.60e−06, 0.0000122]	0.1448
Culture					
PowerDistance (γ_{05})	−0.0016	0.0016104	−1.00	[−0.0047698, 0.0015429]	−0.0381
Individualism (γ_{06})	−0.0018	0.0015728	−1.15	[−0.0023443, 0.0038209]	0.0132
Masculinity (γ_{07})	−0.0004	0.0033915	−0.20	[−0.0073221, 0.0059724]	0.0037
UncertaintyAvoidance (γ_{08})	−0.0016	0.002071	−1.03	[−0.0061873, 0.0019307]	−0.0548
LongTermOrientation (γ_{09})	0.0029*	0.0015448	1.75	[−0.0003267, 0.0057288]	0.0709*
Indulgence (γ_{10})	0.0007	0.0013702	0.49	[−0.0020084, 0.0033626]	−0.0196
Number of effects		199			
Residual heterogeneity (τ^2)		0.0346			
R-squared (%)		6.34%			

Note(s): ***($p < 0.01$); **($p < 0.05$); *($p < 0.1$); ns ($p > 0.1$)
All other statistics (z-value, 95% CI, τ^2, R^2) correspond to the unstandardized model
To account for skewness, *n* (Sample Size) and GDP per capita were log-transformed prior to z-score standardization; all other moderators were standardized using the z-score method

Source(s): Authors' own work

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5. Discussion

5.1 The differentiated effects of various digital technology types on three-stage resilience

Drawing on the interaction between digital technologies and SCR, this research investigates both the "horizontal" and "vertical" dimensions: the former concerns how functional layers such as sensing, analytics and collaboration can be integrated, whereas the latter examines how these functions may yield different outcomes across the "preparedness, response and recovery" stages. Our central theoretical contribution lies in revealing a fundamental asymmetry in the value of digitalization for business process resilience. The findings demonstrate that digital technologies are highly effective at enhancing the "risk perception–dynamic response" loop but are significantly less associated with the more complex "post-disruption reconfiguration–adaptive learning" process. This distinction, quantitatively validated by our meta-analysis, challenges the prevailing view of technology as a uniform solution and instead posits that its value is contingent on the specific process requirements of each resilience stage.

5.1.1 Horizontal dimension of technology. From the "horizontal" perspective, this study classifies digital technologies into information acquisition and sensing technologies, data analytics and decision-making technologies, and collaboration and synchronization technologies, along with multiple/comprehensive technologies. While sensing-layer tools (e.g., RFID, NFC, sensor systems) show relatively modest effect sizes in isolation, their core value lies in providing a front-end interface for data collection, thereby creating foundational visibility throughout the entire supply chain (Taj *et al.*, 2023). When sensing technologies combine with higher-level analysis and coordination tools such as AI, cloud computing or blockchain, raw monitoring data can fuel highly automated and intelligent operational decisions. For instance, AI models can leverage large-scale real-time data gathered at the sensing end to quickly identify abnormal demand or potential disruption risks (Jauhar *et al.*, 2024), significantly boosting the speed and accuracy of organizational responses to disturbances.

Data analytics and decision-making technologies, by harnessing big data and machine learning algorithms for forecasting, scenario simulation and resource allocation, enable organizations to swiftly implement targeted responses. While these intelligent decision systems perform especially well in the "risk perception–dynamic response" domain, they may not comprehensively address the broader human–machine collaboration and flexible management elements needed during post-disruption organizational learning and complex situational assessments.

Collaboration and synchronization technologies further eliminate information silos among enterprises by using distributed mechanisms such as cloud platforms or blockchain to strengthen cross-node process integration (Alikhani *et al.*, 2023). In times of unexpected incidents, such solutions can support synchronized response and resource-sharing among multiple parties, while also serving as a data exchange and business integration foundation in subsequent recovery stages. Particularly in large-scale, networked supply chains, these collaborative platforms – combining real-time communication with traceability functions – can support a more resilient systemic linkage across diverse supply chain partners.

The synergistic benefits of deploying multiple digital technologies arise from structured layering, not simple addition. Industry 4.0, for instance, involves using various technologies together across the value network (Ghobakhloo *et al.*, 2025). Foundational sensing technologies like IoT create a data-rich environment. Analytics technologies like AI use this data to generate predictive insights. Finally, collaboration technologies like blockchain provide a secure mechanism for sharing information between firms, which is essential for coordinated action (Gozali *et al.*, 2024). This layering – where sensing, analytics and collaboration are complementary – builds comprehensive resilience in a way that no single technology can achieve alone.

5.1.2 Vertical dimension of the supply chain. From the "vertical" perspective, this study explores how digital technologies differ in the three major supply chain stages of

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preparedness, response and recovery. First, the preparedness phase is highly reliant on capabilities for risk forecasting and prevention, reflected by its pronounced correlation with data analytics and decision-making technologies (Dai *et al.*, 2025). Early identification of potential threats and the establishment of redundancy strategies not only draw on data from the sensing layer but also rely on intelligent algorithms to support scenario simulation and emergency planning (Ju *et al.*, 2021). If combined with collaboration and synchronization technologies, these capabilities can offer visualized decision-making references for cross-firm or cross-departmental strategic coordination, thus enhancing supply chain stability in the face of risks.

In the response phase when a disruption occurs, platform-based collaboration and real-time data analysis have heightened importance: visibility and instantaneous decision-making allow each link in the chain to adjust resources in tandem, dynamically revise production plans or logistics schemes (Ali *et al.*, 2024). Information acquisition and sensing technologies gather critical inputs at this point, relaying them via collaborative platforms to the decision-making level, facilitating rapid iteration and collective action. Yet as noted regarding the inherent shortfall in "post-disruption reconfiguration–adaptive learning," if algorithms or data analytics focus exclusively on short-term interventions while neglecting deeper analysis of disruption causes or long-range developments, organizations may handle the initial impact effectively but struggle later to advance capability evolution and systemic reconfiguration.

The weaker link between digital technologies and the recovery stage ($r = 0.273$) requires careful interpretation. Our data reveal a critical misalignment: the analytics tools most effective in the preparedness and response stages ($r \approx 0.36$) are the least effective during recovery ($r = 0.256$). This gap is not just about technology; it reflects a fundamental difference in the business processes themselves. Preparedness and response are largely about optimizing activities within established frameworks. In contrast, recovery is about redesign and learning, requiring new routines and models to be created under ambiguous conditions.

This suggests that current data-driven tools, while excellent for optimizing known processes, are less suited for the unstructured problem-solving and organizational learning needed to reconfigure the supply chain after a major disruption. This finding highlights a key limit to the value of current AI and analytics in business process management: they are powerful for executing existing processes but offer limited support for evolving and re-learning them. Instead, blending digital solutions with experiential insights, team-based learning, and supply chain memory can progressively refine recovery strategies and bolster defenses against future similar risks.

The synergistic effect of comprehensive technologies is vividly illustrated in applications such as pandemic vaccine distribution, where integrating IoT for demand sensing with drones for rapid logistics creates resilience across the entire process (Ansari *et al.*, 2025). This moves beyond a simple, stage-isolated view of technology's association with resilience. From a theoretical standpoint, our findings suggest that digitalization is associated with a dynamic, cross-stage process of capability-building. This process can be interpreted through the lens of dynamic capabilities, potentially manifesting as a sequential pathway where digitalization is first linked to supply chain adaptability, which in turn may provide a platform for effective supply chain innovation, ultimately contributing to resilience (Aslam *et al.*, 2025).

Yet precisely because algorithms are relatively more effective for short-term disruption management, deeper systemic transformations required for "post-disruption reconfiguration–adaptive learning" still rely on organizational capabilities and resource alignment to ensure that the resilience advantages unleashed by digital technologies persist and become internalized.

5.2 Influence of moderators and contextual characteristics

This section discusses how the observed relationship between digital technologies and SCR is contextualized by various factors – firm size, industry type, economic development and

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institutional and cultural environments – and explores the potential mechanisms behind the observed moderating effects.

5.2.1 Consistency across firm size and industry characteristics. Regarding differences in firm size and industry, subgroup analysis reveals that although digital technology usage patterns vary across contexts, there is no statistically significant difference in overall effect sizes between the broad categories of manufacturing and services ($p = 0.525$). The underlying reason for this cross-sectoral consistency is that digital technologies primarily augment fundamental business processes – such as demand forecasting, logistics coordination and partner communication – that are ubiquitous across both manufacturing and service supply chains. In high-uncertainty settings like healthcare, for example, data analytics becomes pivotal for managing disruptions (Bag *et al.*, 2024), while in project-based sectors like construction, blockchain is instrumental in fostering the supply chain integration that underpins resilience (Wang *et al.*, 2025). The aggregate non-significant finding therefore reinforces our core thesis: technology's value is realized when it addresses specific, context-dependent vulnerabilities. Similarly, no notable difference emerges between large enterprises and small-to-medium enterprises (SMEs) ($p = 0.219$). These findings challenge earlier assumptions that large firms enjoy a "natural advantage" in technological resources and specialized teams (Qi and Zhang, 2024).

The growing trend of "digital democratization" does not imply that SMEs face no obstacles in digital transformation. On the contrary, implementing digital systems often demands specialized training and upfront infrastructure investments (Jauhar *et al.*, 2024). However, our finding of performance parity suggests that SMEs can compensate for resource limitations with structural agility, enabling a more targeted and rapid integration of select digital tools (Ghobakhloo *et al.*, 2025). This allows them to master internal processes – a key antecedent of SCR (Lin *et al.*, 2025) – by focusing on accessible solutions like No-Code AI rather than large-scale capital projects (Jauhar *et al.*, 2024).

5.2.2 Economic development level and the marginal benefits of digital transformation. Contrasting with the non-significant findings regarding firm size and industry attributes, economic development level exhibits a more pronounced moderating effect. The overall effect size for digital technologies in developing economies ($r = 0.350$) significantly exceeds that of developed economies ($r = 0.267$), and the difference is statistically meaningful ($p = 0.039$) – consistent with findings by He *et al.* (2023) concerning port enterprises. This result suggests that when a country or region has not yet reached the "saturation threshold" of digital infrastructure, digital transformation is associated with more significant marginal returns. For developing economies lacking established supply chain architecture, emerging digital tools frequently provide "latecomer advantages," allowing rapid resource realignment and risk management in the face of extensive disruptions such as trade barriers or geopolitical conflict (Mandal, 2017).

By contrast, developed economies with relatively mature digital ecosystems already incorporate automation and data analytics in routine operations, reducing the "incremental space" in which digital technologies can further improve SCR (Bown, 2023). However, this does not imply a weakness in the digital capabilities of developed countries; rather, it indicates that the next step must involve deeply integrating digital solutions with organizational competencies and pursuing refined management and innovative value-chain development to achieve lasting competitive differentiation (Xue *et al.*, 2024). In this sense, economic development level not only corresponds with the immediate gains from digital transformation but also appears related to how different regions rebuild capabilities and pursue sustainable pathways during a crisis.

5.2.3 Institutional environment: balancing digital dividends with governance mechanisms. The negative moderating effect of the digitalization index ($\gamma = -0.007$, $p < 0.1$) supports our "diminishing digital dividend" hypothesis and signals a critical shift in what drives process resilience. In digitally advanced regions, further technology investments may yield smaller gains. As organizations adopt similar foundational technologies, simply investing more

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without improving management practices can lead to a "digital trap" (Dai *et al.*, 2024). In other words, when technology becomes commonplace, competitive advantage shifts toward better institutional and governance capabilities (Lin *et al.*, 2025).

From a governance perspective, this statistical trend can be interpreted as evidence of a growing imbalance: a gap between "technology excess" and "inadequate governance," where digital expansion outpaces the cross-organizational coordination and risk-control mechanisms needed to harness its full potential (Li *et al.*, 2024; Zhu *et al.*, 2023). This suggests that to counteract diminishing returns, the focus of process improvement may shift from firm-level technology to inter-firm governance. This requires a synergy where digital contractual governance (e.g., via blockchain) creates a transparent, enforceable foundation of rules (Wang *et al.*, 2025), which in turn enables the trust-based relational governance needed for adaptive, collaborative problem-solving across the ecosystem (Lu *et al.*, 2023; Peng *et al.*, 2024).

5.2.4 Cultural perspective: cross-national variations and trust mechanisms. Macro-level cultural differences add further explanatory depth to how digital technologies enable SCR (Hofstede, 2001). Although only long-term orientation (LTO) exhibits a significant effect in this study, while other cultural measures such as power distance or uncertainty avoidance do not, this does not necessarily mean they lack relevance. Instead, more complex or nonlinear interplays may be at work (Dubey *et al.*, 2020). In this study, higher LTO tendencies encourage sustained investments in digital infrastructure and risk management, leading to quicker recovery from disruptions. In contrast, cultural traits such as high power distance or strong uncertainty avoidance may support rapid centralization and consistent directives in the initial stages of a crisis but can create organizational rigidity during subsequent collaboration and innovation.

Trust is another critical factor. Cross-national supply chains routinely involve diverse cultural settings. In environments marked by higher trust levels and stronger collaborative norms, organizations are typically more inclined to use digital platforms for information-sharing and collective decision-making, thereby mitigating asymmetrical information risks. Conversely, in contexts emphasizing individualism or lacking trust in collaboration, more stringent governance mechanisms (e.g. blockchain or smart contracts) may be required to prevent moral hazard and alleviate anxieties about partnership (Hofstede, 2001). Notably, lower power distance and greater indulgence can enhance the fluidity of internal communication, encourage team collaboration and increase risk tolerance (Alsanoosy *et al.*, 2019). This cultural backdrop enables digital technology to permeate all supply chain stages more freely, bolstering overall agility and synchronization across the network.

In summary, our analysis suggests that cultural factors are profoundly intertwined with how organizations perceive and adopt digital solutions, as well as with the governance frameworks and resource allocation decisions for handling uncertainty. For future research, applying cross-cultural perspectives and multi-case longitudinal studies could yield a deeper understanding of how cultural dimensions interact with digital technologies to augment SCR.

6. Conclusion, implications and limitations

6.1 Main conclusions

This study addresses the four key research questions raised in the introduction. First, through an extensive literature review, digital technologies were categorized into three core functional categories. Then, employing a meta-analysis approach, we systematically examined the differentiated associations between digital technologies and the three phases of SCR. The results indicate that digital technologies are significantly associated with resilience in the preparedness ($r = 0.332$), response ($r = 0.322$) and recovery ($r = 0.273$) stages, yet these associations are uneven across the phases. Data analytics and decision-making technologies show the strongest association during preparedness ($r = 0.361$) and response ($r = 0.366$), whereas the recovery stage is more strongly linked to information acquisition and sensing technologies ($r = 0.299$). This finding suggests that digital technologies are more strongly

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associated with the "risk perception–dynamic response" loop than with the "post-disruption reconfiguration–adaptive learning" process. Additionally, the study confirms the significant moderating effects of macro-level variables: the association is stronger in developing economies ($r = 0.350$) than in developed economies ($r = 0.267$), while the digitalization index ($\gamma = -0.007$) acts as a negative moderator and a long-term orientation culture ($\gamma = 0.003$) serves as a positive moderator. These findings highlight that the phenomenon of diminishing digital dividends and cultural values are strongly intertwined with digitalized pathways toward building SCR.

6.2 Theoretical contributions

This study makes three primary contributions to the literature on SCR and digitalization. First, it develops an innovative cross-stage analytical framework that integrates the dynamic process view of SCR with the multilayer functional perspective of digital technologies, thus answering Ponomarov and Holcomb's (2009) call to examine the multi-stage dynamics of resilience. Second, our large-scale quantitative analysis uncovers the "uneven effect" of digital technologies. We show their effectiveness is significantly higher for risk prevention and immediate response than it is for rebuilding systems and evolving capabilities after a disruption. This finding extends the research boundary of Spieske and Birkel (2021). Third, by incorporating macro-level moderating variables, the study demonstrates that economic development levels, digital maturity and cultural values collectively shape how digital technologies enable resilience, notably revealing the phenomenon of "diminishing digital dividends" and the positive influence of long-term orientation culture on digital transformation. These findings enrich social and cultural considerations in research exploring digital technologies' support for supply chains (Kumar and Raj, 2025).

6.3 Managerial implications

The study offers insights on three levels for enterprise managers and policymakers. At the technology planning level, firms should adopt differentiated, evidence-based deployment strategies aligned with our stage-specific findings. For the preparedness ($r = 0.332$) and response ($r = 0.322$) stages, investments should prioritize analytics and decision-making technologies, which demonstrate the strongest effects ($r \approx 0.36$). Conversely, for the recovery stage – where our findings reveal analytics are less effective ($r = 0.256$) – the strategy must pivot to integrating sensing technologies ($r = 0.299$) with organizational learning to address the critical "post-disruption reconfiguration–adaptive learning" gap.

At the organizational structure level, firms must build decision-making frameworks that align with digital technologies. For SMEs, our finding of performance parity with large firms ($p = 0.219$) suggests a strategy of prioritizing mastery over breadth: achieving deep process integration with a few targeted, low-cost digital tools (e.g. a cloud-based logistics platform) rather than attempting a large-scale, resource-intensive transformation.

Finally, at the ecosystem level, our moderator analysis provides targeted policy guidance. For developing economies, where technology yields higher returns ($r = 0.350$), infrastructure investment remains the priority. In contrast, developed economies experiencing diminishing returns ($\gamma = -0.007$) should redirect resources toward establishing governance frameworks and institutional standards for data sharing and automated decision-making. Additionally, societies with strong long-term orientation ($\gamma = 0.003$) are better positioned to frame digital investments as strategic foundations for future competitiveness, suggesting more favorable conditions for sustained technology adoption policies.

6.4 Limitations and future research directions

Although this study broadens the understanding of the relationship between digital technologies and SCR, several limitations remain. First, due to the present sample

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conditions, it is not possible to measure correlations among the three stages of SCR within the current research design. Second, the study falls short of fully capturing the dynamic interactions among different types of digital technologies.

This paper's findings open several promising avenues for future research that can build directly upon our contributions. First, our discovery of an "uneven effect" across resilience stages calls for mixed-methods studies to uncover the micro-foundations of this quantitative pattern; specifically, longitudinal case research is needed to explore how business processes and digital tools co-evolve through the preparedness, response and recovery phases. Second, our results reveal a critical "capability misalignment": the analytics tools effective in early stages are weakest during recovery. Future inquiry should therefore investigate how to design systems that effectively blend technology with organizational routines and human skills to support the unstructured, learning-intensive processes that define recovery. Finally, our findings on the "diminishing digital dividend" in mature economies and the retrospective nature of our analysis highlight the need for future-oriented research. The next step is to evaluate whether emerging technologies like generative AI can overcome the capability misalignment we identified and offer new pathways to resilience in an increasingly complex global landscape.

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Supplementary material

The supplementary material for this article can be found online.

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